Test Method for Ultraviolet Light Discoloration	Test method #: VF- E03 REF #: TM-R- 058 / TM-R-069 Page 1 of 4
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PURPOSE AND SCOPE: Option 1: To measure the tendency of the surface of a material to discolor as a result of exposure to ultraviolet light.

Option 2: To describes a technique for evaluating surface discoloration of leather due to exposure to ultraviolet radiation.

REFERENCE: ASTM-D 1148-95

EQUIPMENT AND MATERIAL:

1. Ultraviolet light oven with revolving pan, capable of maintaining 50± 1°C (Option1)
2. OSRAM Ultra –Vitalux or similar 300watt UV light bulb – 1 unit (Option 1)
3. N- UV light test chamber - Qualitest GT-7035-UB Discoloration Meter (Sankyo Denki, G15T8-15W), or similar (Option 2)
4. 15W UV bulbs – 2 units (Option 2)
5. Cutting dies: 6 cm x 3 cm (Option 1) and 5.7cm x 8.9cm (Option 2)
6. Thermometer, capable of reading to 0.5◦C
7. Masking tape, non-transparent
8. White clean paper to cover the specimen
9. AATCC Chromatic Transference Scale or gray scale
10. Light box with D65 light
11. Fabric, Leather, rubber, plastic, synthetic leather and adhesive systems.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

2. Die or hand cut the test sample 6 cm long *3 cm wide if possible the thickness of the sample should be 5.0mm.

EQUIPMENT PREPARATION:

1. Adjust the rotating table such that it is 250 ±2 mm below the bottom of the ultralight source.

2. Set the speed of the rotating table to 3 ± 1 revolution per minute.

REPETITIONS: Test two (2) samples per material.

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TEST PROCEDURE: Option 1

1. Mask the first and last 2cm of the test sample with a piece of paper at least 4 layers, then fasten with tape but ensure that the tape should not touch the test sample. This will leave the center 2 cm unmasked.
2. Place the test sample on a rotating table such that the closest edge is 6.0 ± 0.5 cm from the center of the table. (The length of the test sample must be parallel to the radius of the rotating table) See figure A.

3. Expose the test sample to the ultraviolet ray for 12 hours or 24 hours at a temperature of 50±1 °C. (The temperature should be verified by placing a reliable thermometer adjacent to the test sample.)

4. Remove the test sample from the chamber and then remove the masking tape from the test sample.

6. Condition the sample for 1 hour.

7. Place the test specimen in the light box, set for 'Daylight' (D65). Examine the change in color - compare the color of the darkest portion of the uncovered area of the test sample. A discoloration rating shall be assigned based upon the AATCC Chromatic Transference Scale / AATCC Gray Scale for Color Change.

Note:

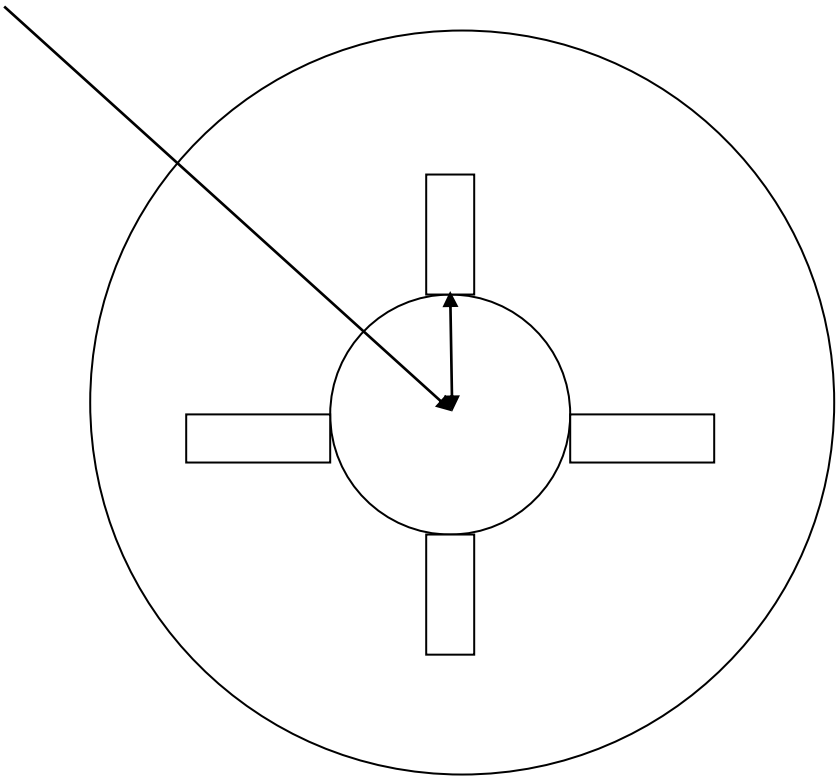
- Chromatic Transference Scales must be replaced not beyond ***12 months.***
- Light bulb must be replaced at least every 1000 hours actual usage. ***(Total number of hours will be calculated by maintaining a test log wherein every test is recorded and add every test until the maximum 1000 # of hours is reached. Alternatively, the test chamber can be equipped with an hour meter that logs total hours of usage.***

REPORT: The ultraviolet light discoloration to the nearest 0.5unit base on the AATCC Chromatic Transference Scale / AATCC Gray Scale. **Discoloration rating 1-5 AATCC scale.**

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Figure A:

The closest edge is 6.0 ± 0.5 cm from the center of the rotating table.



TEST PROCEDURE: Option 2

1. The test specimen shall be prepared from the leather sample. The test specimen shall be rectangular in shape, 5.7cm x 8.9cm. For production parts, it is acceptable to use a different specimen size. The actual specimen size shall be noted in the final report.

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2. Adjust the height of the shelf so that the bottom of the bulb is 250mm ± 10mm (10in. ± 0.5in.) above the surface of the test specimen.
3. Place the specimen(s) on the shelf with the surface to be tested facing the UV bulbs.
4. Mask one half of the surface to be tested with a piece of paper at least 4 layers, then fasten with tape to prevent exposure to the UV light but ensure that the tape should not touch the test sample
5. Expose the test specimen(s) to the ultraviolet light for a period of 24 hours. Both bulbs should be used for a total of 30W.
6. At the end of the 24hour period, remove the test specimen from the test chamber.
7. Condition the sample for at least 1 hour.
8. Place the test specimen in the light box, set for 'Daylight' (D65), and examine the change in color. A discoloration rating shall be assigned based upon the AATCC Gray Scale for Color Change.

NOTE: The bulb shall be replaced when the total hours of usage reaches 500 hours.

Bulb replacement timing will be calculated by maintaining a test log where the number of tests is recorded and multiplied by the hours required for each test. Alternatively, the test chamber can be equipped with an hour meter that logs total hours of usage.

REPORT: Discoloration rating 1-5 AATCC scale.

Note: Include - material description, type of bulb used, distance from the bulb to the test specimen, exposure time in hours, size and shape of specimen if not at the required dimension of 5.7cm x 8.9cm, and any deviations from this test method.